# Nonparametric Analysis of Feature-Driven Performance in Wearable Heath Devices

Ester Maiocchi (10800862), Marianna Mazza (10838595), Arianna Perotti (10811649)

2026-02-17

## Contents

# Introduction

## Dataset presentation

The wearable health device market has experienced extraordinary global growth in recent years and is expected to continue expanding rapidly. According to industry forecasts, this market is projected to grow from approximately USD 35.3 billion in 2025 to over USD 144 billion by 2035, representing a compound annual growth rate of more than 15% over the next decade. This expansion is driven by heightened consumer interest in proactive health and fitness monitoring, increasing adoption of digital health solutions and the integration of advanced technologies such as artificial intelligence and real-time biometric analytics into wearable devices . As individuals become more focused on personalized wellness and preventative care, wearables are rapidly transitioning from niche gadgets to essential tools for everyday health management, spanning applications from activity tracking to sleep analysis and stress monitoring. The big size and momentum of this market underline the importance of understanding not just how these devices perform technically, but also how their price, features and brand positioning translate into user satisfaction and consumer value. [market.us, Feb 2026].

The dataset we will use for this analysis is a collection of detailed information on a broad selection of wearable health devices, downloaded from Kaggle [Kaggle, 2025]. It includes products from leading manufacturers and spans multiple device categories, such as smartwatches, fitness trackers, smart rings, fitness bands and sports watches. Consistency between the market overview and the empirical analysis is ensured since all the devices in our dataset are mentioned in articles that discuss this topic [Rachael Schultz, Steven Cohen, 2026]. This variety allows for a comprehensive comparison across heterogeneous devices that differ in intended use, technological sophistication and target users.

The dataset has 2375 devices recording both technical characteristics and performance-related variables.

| Variables | Description |
|---|---|
| Test_Date | Date of performance evaluation (YYYY-MM-DD) |
| Device_Name | Full name of the device (Brand + Model) |
| Brand | Manufacturer of the device (Apple, Samsung, Garmin, Fitbit, Huawei, Amazfit, Whoop, Oura, Polar, Withings) |
| Model | Specific model name |
| Category | Device type |
| Price_USD | Retail price in US dollars |
| Battery_Life_Hours | Battery life (in hours) on a full charge |
| Heart_Rate_Accuracy_Percent | Heart rate monitoring accuracy (%) |
| Step_Count_Accuracy_Percent | Step counting accuracy (%) |
| Sleep_Tracking_Accuracy_Percent | Sleep tracking accuracy (%) |
| Water_Resistance_Rating | Water resistance certification (IPX4, 3ATM, IPX7, 5ATM, IPX8, IP68, 10ATM) |
| User_Satisfaction_Rating | User satisfaction score (1–10 scale) |
| GPS_Accuracy_Meters | GPS accuracy (in meters, if available) |
| Connectivity_Features | Supported connectivity options (Bluetooth, WiFi, NFC, LTE) |
| Health_Sensors_Count | Number of health-related sensors |
| App_Ecosystem_Support | Supported app ecosystems (iOS, Android, Cross-platform) |
| Performance_Score | Composite performance score (0–100) |

The coexistence of objective technical specifications and subjective evaluation metrics makes the dataset particularly suitable for studying the relationship between design choices, technological complexity and perceived performance. Its cross-sectional nature is designed to reflect the current state of the wearable health device market in the United States.

## The Stakeholder

The stakeholder considered in this study is the customer interested in purchasing a wearable health device. In an increasingly saturated market, potential buyers are faced with a wide range of products that differ substantially in price, technical specifications and advertised performance. For a typical user, the purchasing decision is rarely driven by a single characteristic; instead, it involves evaluating a combination of accuracy, battery life, available features, comfort and overall usability. Customers are therefore interested in identifying which category of wearable devices offers the most balanced compromise among these dimensions, ensuring reliable health monitoring without unnecessary complexity or excessive cost.

A central concern for consumers is whether the price of a device reflects genuine improvements in accuracy and functionality or is largely influenced by brand popularity and market positioning. Well-established brands often command higher prices, but it is not always clear whether this premium corresponds to measurable gains in performance or user satisfaction. From the customer's perspective, an objective, data-driven assessment of the relationship between price, technical characteristics and perceived quality is essential to make informed purchasing decisions and to avoid overpaying for features that may offer limited practical benefit.

## Research Questions

In light of the customer-centered perspective outlined above, here are our research questions:

1. **Which device category offers the best balance of accuracy, battery life and other features for typical users?**

   It seeks to determine which category of wearable health devices provides the best balance between accuracy, battery life and additional features for a typical user. This analysis aims to highlight potential trade-offs across device types and to identify categories that deliver robust performance without excessive compromises.

2. **Is a higher price justified by better accuracy and functionality or just by the brand's popularity?**

   Here we aim to investigate whether higher-priced devices are justified by superior accuracy and functionality or whether price differences are primarily driven by brand reputation rather than objective performance metrics. By examining the association between price, sensor characteristics and performance scores, the analysis aims to separate the role of technological quality from branding effects.

3. **Can we predict how a new device would perform in terms of user satisfaction?**

   We want to explore whether it is possible to predict the performance of a new wearable device in terms of user satisfaction based on its observable characteristics. This predictive perspective is particularly relevant for customers evaluating newly released products with limited user feedback, as well as for understanding which features are most influential in shaping overall satisfaction.

# Analysis

## Data preprocessing

Before conducting the statistical analysis, a preprocessing phase was carried out to ensure consistency and usability of the dataset. The raw data were first inspected to assess the distribution of devices across brands and categories, which allowed for an initial understanding of the market representation as shown in Figure 1.

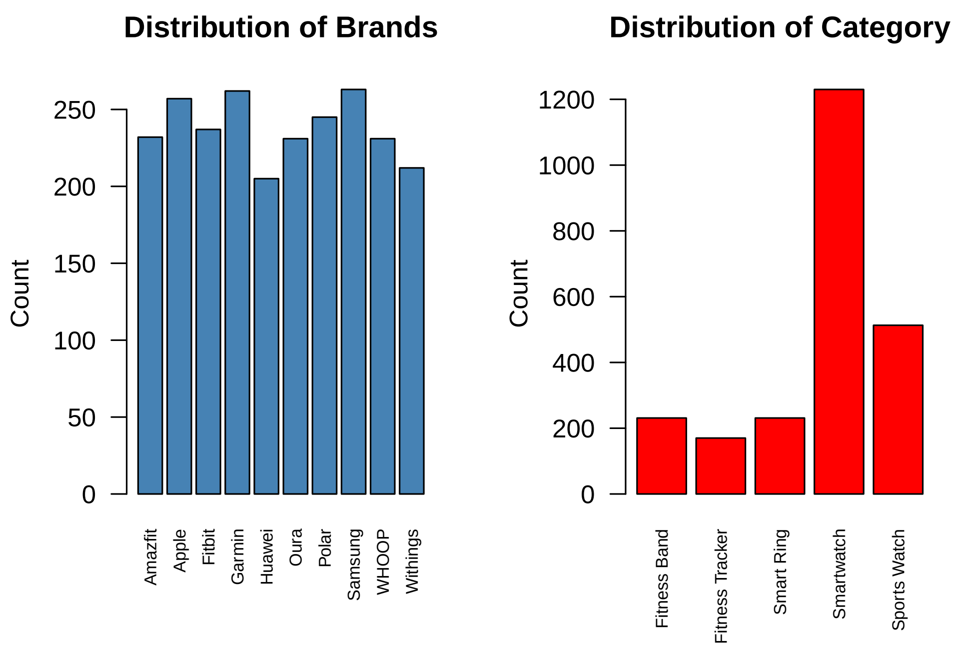


Figure 1: Distribution of brands and categories

Missing values in the GPS accuracy variable were interpreted as the absence of GPS functionality; to retain these devices in the analysis and make this information explicit, missing GPS accuracy values were replaced with a large numerical value representing very low accuracy.

Categorical variables describing the device identity, brand, model and category were converted into factors to enable appropriate handling in subsequent analysis. In addition, ordinal features such as water resistance, connectivity options and app ecosystem support were transformed into ordered factors, reflecting their natural progression in terms of technological sophistication.

## RQ1: Which device category offers the best balance of accuracy, battery life and other features for typical users?

To address the first research question, a composite balance index was constructed to capture the overall trade-off between accuracy, battery life and technological features at the device level. Several key performance dimensions were considered, including heart rate, step count and sleep tracking accuracy, battery life, GPS precision, water resistance, connectivity, app ecosystem support and the number of health sensors. Each variable was transformed into a rank, with lower ranks assigned to better-performing devices, while GPS accuracy was ranked in the opposite direction to reflect lower error as better performance. The balance index was then defined as the average of these ranks, providing a global, nonparametric measure in which lower values correspond to a more favorable balance of features.

Figure 2 presents the distribution of the balance rank across device categories.

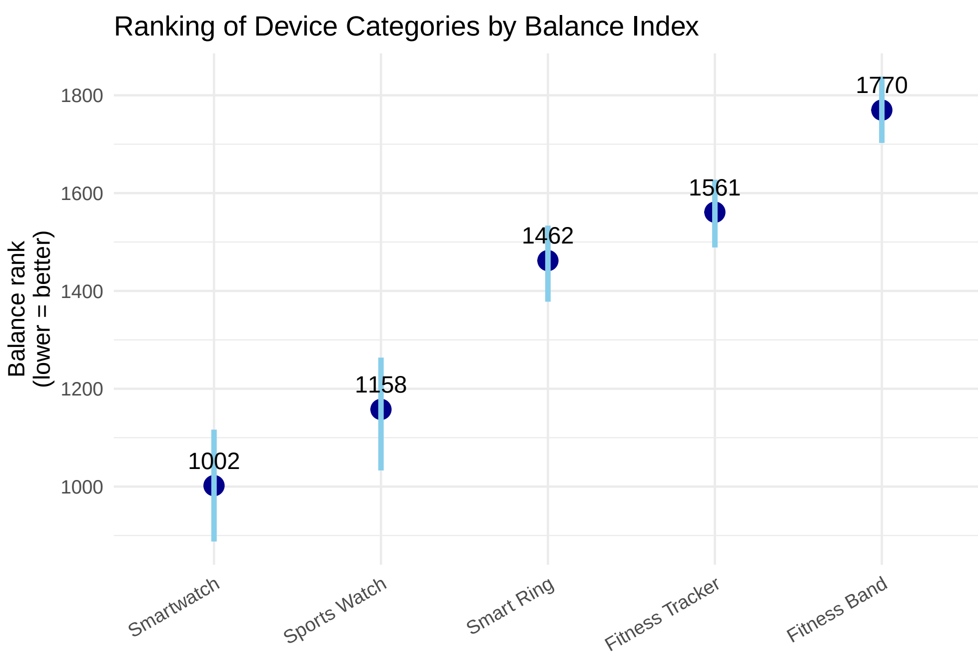


Figure 2: Ranking of device categories by balance index

Smartwatches exhibit the lowest median balance rank, indicating that, on average, they provide the most favorable combination of accuracy, battery life and features among the considered devices.

Sports Watches rank second overall, with a slightly higher median balance rank and a wider variability. This reflects the coexistence of highly specialized, high-performance devices alongside models that prioritize specific functionalities at the expense of others.

Smart Rings occupy an intermediate position, showing moderate balance ranks but also a relatively narrow spread, indicating homogeneous performance across models.

In contrast, Fitness Trackers and Fitness Bands display higher median balance ranks, implying a less favorable balance of features. Fitness Bands, in particular, appear to perform worst overall, with both higher central values and limited variability, consistent with their more basic design and reduced sensor and feature sets.

The formal differences of the balance ranks were assessed through a permutation-based one-way ANOVA. Category turned out to be a significant factor, so there is at least one category that has a different overall performance from the others.

Significant global differences motivated pairwise permutation tests on category medians. For each pair of categories, the absolute difference between sample medians was used as the test statistic and its significance was assessed through permutation under the null hypothesis of equal distributions. All pairwise contrasts resulted in p-values equal to zero. This indicates that the median balance rank differs significantly between every pair of device categories, providing strong evidence that each category represents a distinct performance profile in terms of the overall balance between accuracy, battery life and features.

Given the prominent role of Smartwatches and Sports Watches in the ranking and in everyday life, a more detailed comparison between these two categories was subsequently performed.

For continuous performance metrics, Wilcoxon–Mann–Whitney tests were used to assess distributional differences.

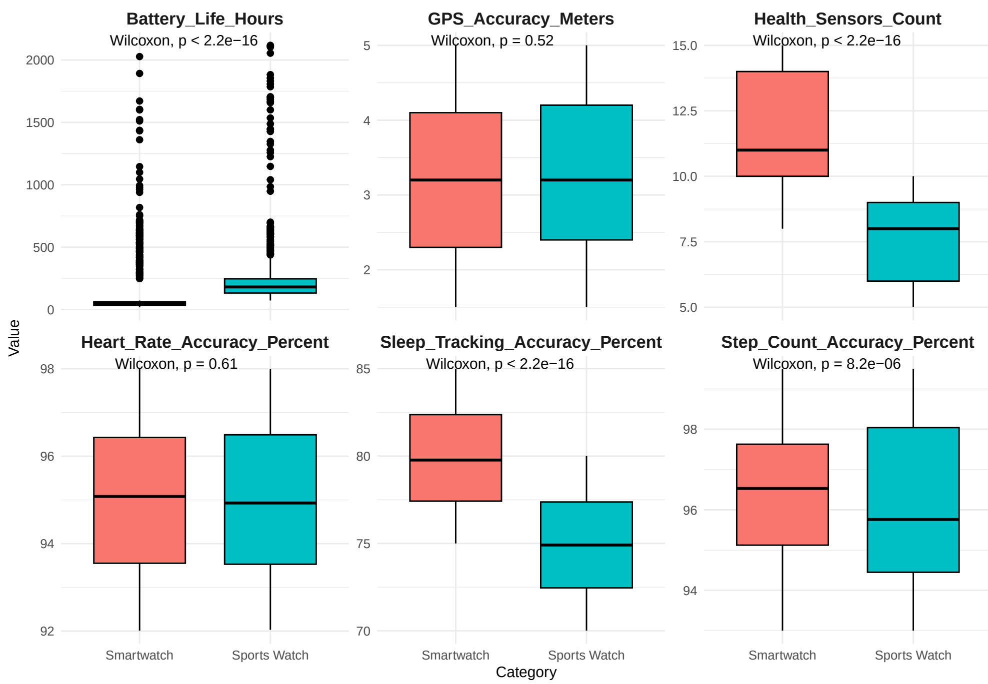


Figure 3: Comparisons between continuous variables of Smartwatches and Sport watches

As shown in Figure 3 clear differences emerge between the two categories, highlighting distinct design priorities. Sports Watches exhibit a substantially longer battery life. On the other hand, Smartwatches tend to include a larger number of health sensors and a more accurate tracking of sleep and steps. By contrast, GPS and heart rate accuracy do not differ significantly between the two categories.

Then, categorical features such as water resistance, connectivity options and app ecosystem support were analyzed using permutation-based chi-square tests to evaluate whether their distributions differed significantly between the two categories.

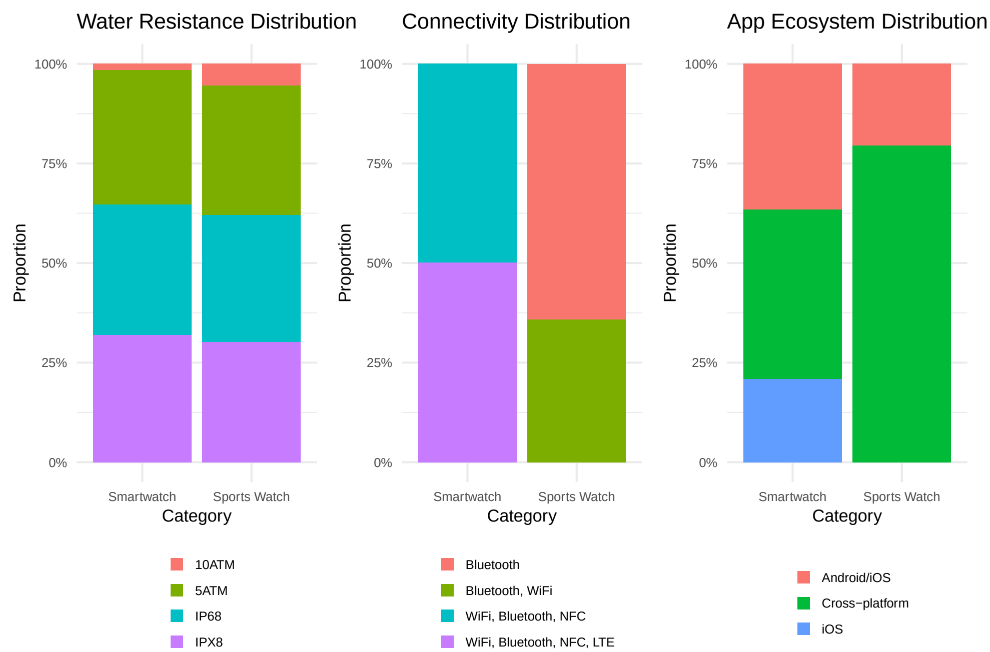


Figure 4: Comparisons between categorical variables of Smartwatches and Sport watches

Figure 4 shows that water resistance of most devices is IP68, IPX8 or 5ATM, with a smaller proportion rated 10ATM. Even though in the plot the distribution seems fairly similar, the test reveals that there is a significant difference between the two categories.

In the Connectivity plot, Smartwatches tend to support multiple connectivity options (WiFi, Bluetooth, NFC, LTE), while Sports Watches are dominated by simpler connectivity (Bluetooth or Bluetooth, WiFi). In fact, the test highlights this substantial difference.

The App Ecosystem plot shows that Sports Watches are largely cross-platform, whereas Smartwatches are more evenly distributed among iOS, Android/iOS and cross-platform support. Notably, the presence of Apple devices in the Smartwatch category limits compatibility, reducing the proportion of cross-platform options.

Overall, Smartwatches emerge as the category offering the best balance of accuracy, battery life and technological features for typical users, closely followed by Sports Watches. While Sports Watches excel in battery longevity and app ecosystem support, Smartwatches provide more comprehensive sensor coverage, better step and sleep tracking and broader connectivity. These results highlight that device categories satisfy distinct user priorities, with Smartwatches favoring multifunctionality and Sports Watches emphasizing endurance and specialized performance.

### RQ2: Is a higher price justified by better accuracy and functionality or just by the brand's popularity?

To address the second research question, we first constructed a synthetic measure of device quality that captures accuracy and functionality independently of price and brand, rather than relying directly on the performance score provided in the dataset. The original performance score may implicitly incorporate subjective or marketing-related considerations and its construction is not transparent, making it difficult to disentangle pure technical quality from brand positioning effects. This concern is further supported by the observed correlations between the performance score and individual technical variables: for instance, the score is strongly positively correlated with GPS accuracy and negatively correlated with heart rate accuracy, indicating that higher values of the score do not consistently correspond to better technical performance across metrics. This lack of coherent alignment with objective accuracy measures makes the interpretation of the original score ambiguous. Since the aim of this research question is to assess whether higher prices are justified by genuine improvements in accuracy and functionality, relying on such an internally inconsistent measure would compromise the validity of the analysis.

Starting from the original technical specifications, we transformed all performance-related variables into normalized rank scores, so that each feature contributes on a comparable, scale-free basis. In particular, GPS accuracy was inverted to ensure a coherent interpretation where higher values always correspond to better performance. These ranked features were then aggregated into two latent components: an Accuracy Index, summarizing tracking precision across heart rate, steps, sleep, and GPS; a Functionality Index, capturing usability-related aspects such as battery life, number of sensors, water resistance, connectivity and app ecosystem support. The two components were combined with equal weights into a single Performance Index, representing an overall measure of technical quality.

We excluded Fitness Band from our analysis because every device in this category has the same price.

To formally assess whether higher prices are justified by better technical performance or are instead driven by brand effects, we modeled device price as a function of the constructed Performance Index and brand identity using a generalized additive model. The GAM framework allows us to capture potential nonlinear relationships between price and performance while simultaneously accounting for systematic brand-specific price shifts through parametric effects, as follows:

$$\text{Price}_i = \beta_0 + f(\text{Performance Index}_i) + \sum_{b=1}^{8} \beta_b \, I\{\text{Brand}_i = b\} + \varepsilon_i$$

where $i$ is the device index.

When both performance and brand are included in the model, the smooth effect of the Performance Index is not statistically significant, whereas brand effects are large and highly significant for almost all manufacturers. This suggests that, once brand identity is taken into account, differences in accuracy and functionality do not explain a substantial share of price variability. In contrast, brand coefficients indicate significant price increases, particularly for well-established brands such as Apple and Garmin, whose estimated effects correspond to several hundred dollars above the baseline price level.

Figure 5 illustrates the estimated partial effect of technical performance on device pricing within the GAM framework. The solid line represents the estimated spline function, which appears relatively flat and centered near zero across the majority of the performance spectrum, only exhibiting a slight upward trend at the highest index levels. The dashed lines representing the 95% confidence intervals remain wide and encompass the zero horizontal axis throughout the entire range of the covariate. This visual evidence confirms the lack of statistical significance reported in the model output, indicating that the Performance Index does not exert a reliable, systematic influence on price.

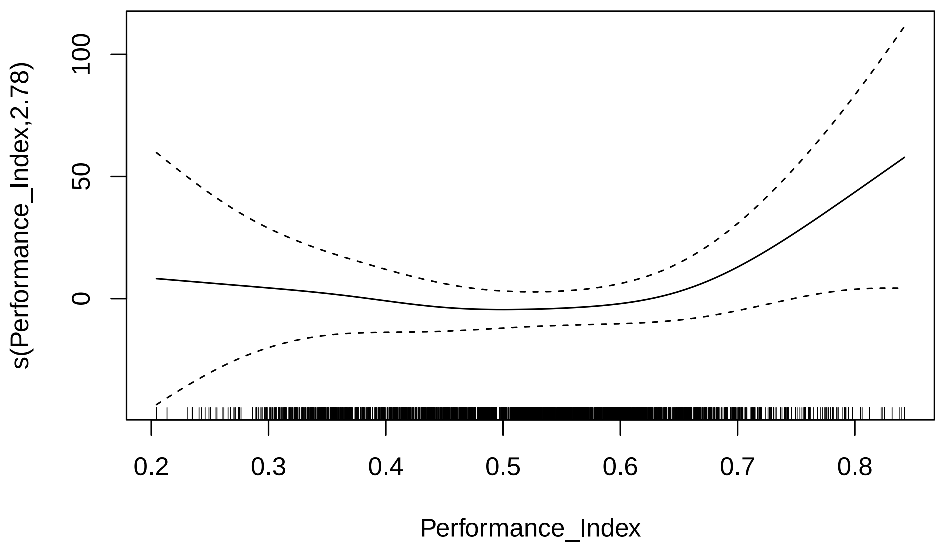


Figure 5: Smooth plot of Performance Index

Residual diagnostics do not indicate severe departures from normality (Figure 6), suggesting that the Gaussian assumption is reasonably adequate for the fitted model. This supports the validity of the inference. Moreover, the overall explanatory power of the model is high when brand information is included, with nearly 44% of the price variability explained.

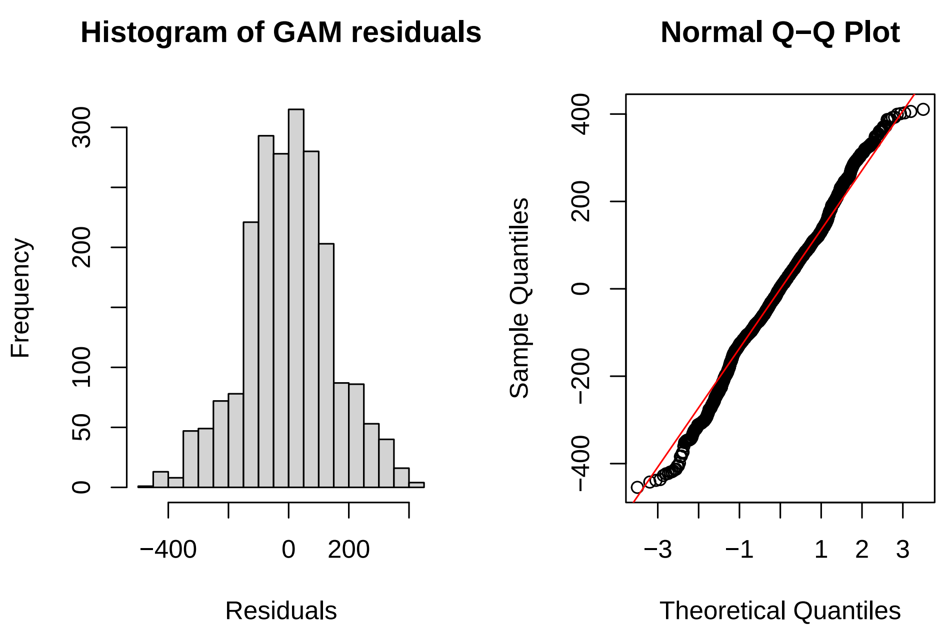


Figure 6: Residuals of the full GAM

To isolate the role of technical performance alone, we also fitted a GAM without brand effects, in the following way:

$$\text{Price}_i = \beta_0 + f(\text{Performance Index}_i) + \varepsilon_i$$

where $i$ is the device index.

In this case, the Performance Index exhibits a strong and highly significant nonlinear association with price, but the proportion of explained variability drops dramatically to around 12%. This indicates that while better accuracy and functionality are associated with higher prices in isolation, they account for only a limited fraction of price differences across devices. The comparison between the two models therefore provides clear evidence that a large portion of price variation is attributable to brand-related factors rather than to objective improvements in performance.

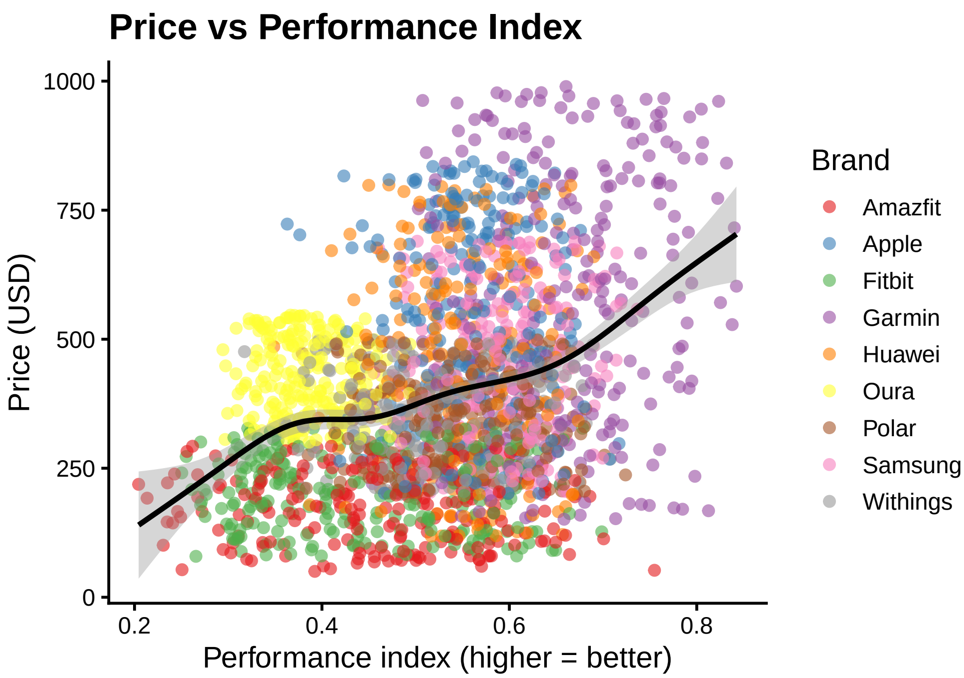


Figure 7: Price vs Performance Index coloured by Brand

Figure 7 represents the univariate relationship between technical performance and price, corresponding to the model without brand effects. This visualization highlights a clear, monotonic upward trend: as the Performance Index increases, the average price of the device also tends to rise.

However, a closer inspection of the data points, colored by brand, reveals the underlying reason for the low explained variability (12%). While the general trend is upward, there is massive vertical dispersion at any given point along the performance axis. For instance, at a Performance Index of approximately 0.5, prices range from nearly \$100 to almost \$1000. Brands like Garmin and Apple consistently cluster in the higher price brackets regardless of their specific performance score, while brands like Amazfit and Fitbit occupy the lower tier. This visual evidence confirms that while a “baseline” performance-price relationship exists, the vast majority of the price premium is dictated by brand positioning rather than technical specifications.

To further investigate the relationship between technical performance and price, we complemented the mean-based analysis with a quantile regression approach, which allows the effect of the Performance Index to vary across different points of the price distribution. Unlike standard regression models that focus exclusively on the conditional mean, quantile regression provides a more comprehensive description of the conditional distribution of price, making it particularly suitable in the presence of heteroskedasticity and asymmetric dispersion, both of which are evident in the price–performance relationship, as shown in Figure 8.

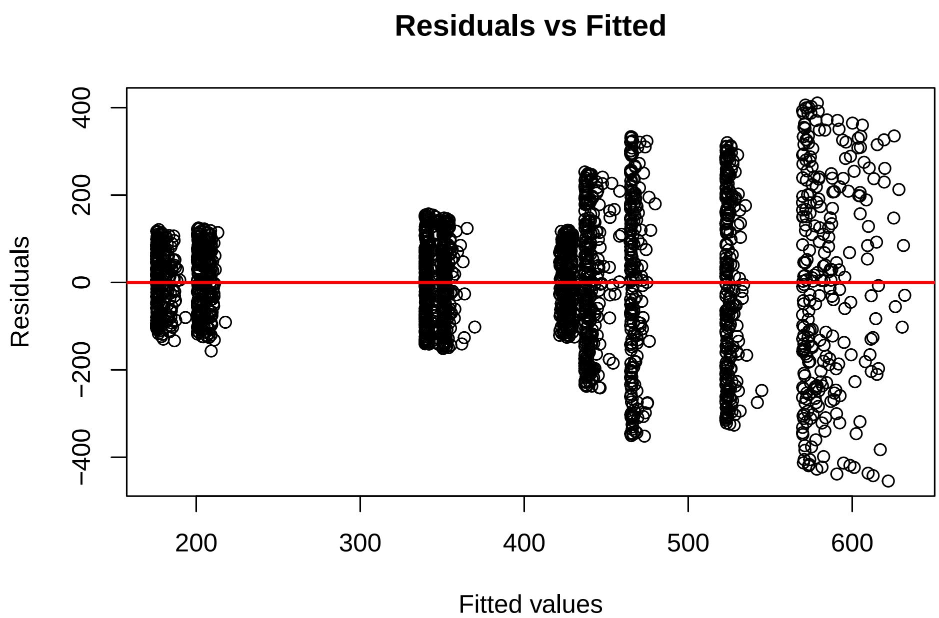


Figure 8: Heteroscedasticity of residuals

We estimated linear quantile regressions for multiple quantiles ranging from the lower tail (0.05) to the upper tail (0.95) of the price distribution. This approach enables us to assess whether improvements in accuracy and functionality translate into higher prices uniformly across the market, or whether their impact differs between low-priced and high-priced devices. The fitted quantile coefficients in Figure 9 reveal substantial variation in slopes across quantiles, indicating that the association between performance and price is not constant.

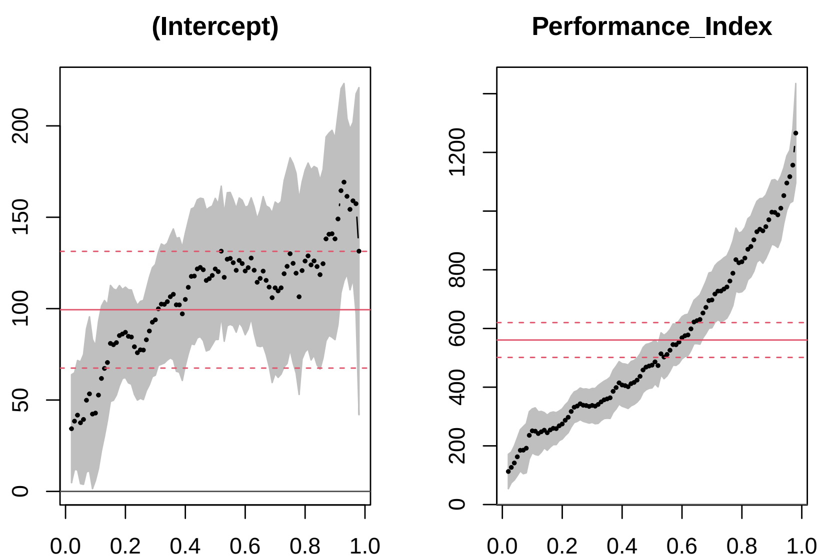


Figure 9: QR Coefficients for Price vs. Performance

Looking at Figure 10, at the lower quantiles of the price distribution, represented by the 0.05 and 0.25 lines, the slope with respect to the Performance Index is relatively mild. This suggests that among low-priced devices, improvements in accuracy and functionality translate into only modest price increases.

Moving toward the median, the relationship becomes more pronounced, indicating that performance plays a clearer role in pricing decisions in the central part of the market.

The effect is strongest at the upper quantiles, particularly for the 0.75 and 0.95 quantiles, where the slopes are steeper and the separation between quantile lines increases. This implies that in the high-price segment, higher technical performance is more strongly associated with higher prices, but also that price dispersion remains large even among high-performing devices.

As a result, this analysis provides additional evidence that technical performance alone cannot fully account for price variability, especially in the lower and middle segments of the market, reinforcing the conclusion that brand-related factors contribute substantially to price formation.

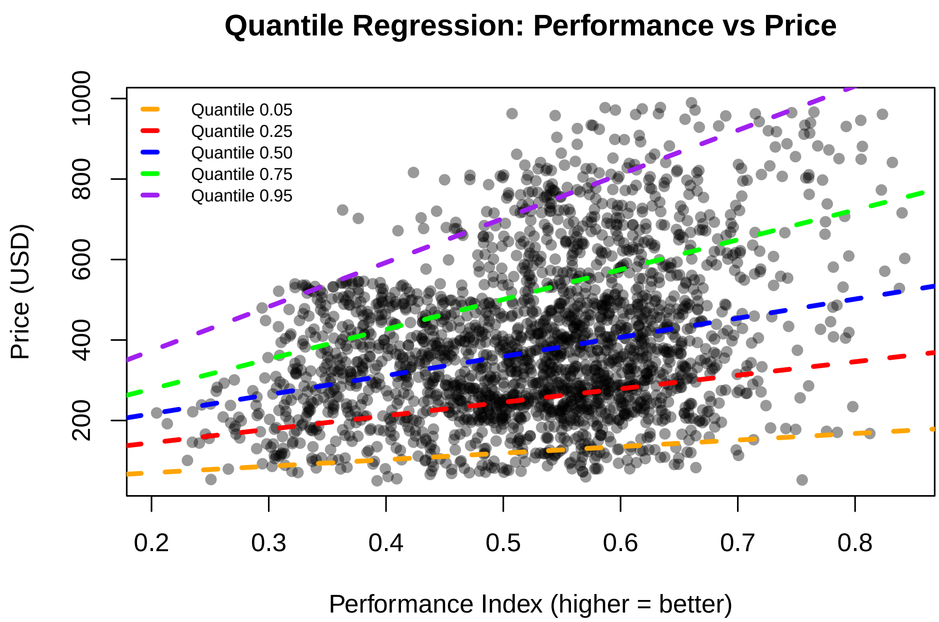


Figure 10: Quantile regression

Overall, the combined evidence from the GAM and quantile regression analysis indicates that while technical accuracy and functionality do play a role in shaping device prices, their explanatory power is limited once brand identity is taken into account. Price differences in the market are therefore largely driven by brand positioning and perceived brand value rather than by proportional improvements in objective performance, especially in the low- and mid-price segments.

### RQ3: Can we predict how a new device would perform in terms of user satisfaction?

Now, we shift the focus from explaining observed prices to predicting the expected level of user satisfaction for a new, unseen device, while explicitly quantifying the uncertainty associated with such a prediction. Rather than producing a single point estimate, the goal here is to construct prediction intervals that are valid in a distribution-free sense, providing a good answer to how well a new device might be rated by users.

To this end, we adopted a conformal prediction framework applied directly to the distribution of the user satisfaction ratings. The idea is to evaluate how plausible a hypothetical satisfaction value is when compared to the empirical distribution of the observed ratings.

Several nonconformity measures are considered to assess robustness. First, a mean-based absolute deviation (T-type) measure captures how far a candidate rating deviates from the central tendency of the data. Then, a k-nearest-neighbor distance, which is sensitive to local data density and thus adapts to possible multimodality or skewness in the satisfaction distribution. Finally, a Mahalanobis-distance-based measure is used to account for the global dispersion structure of the data. By comparing the resulting conformal prediction intervals across these different choices (Figure 11), we obtain a comprehensive assessment of the range of user satisfaction values that are statistically consistent with past observations.

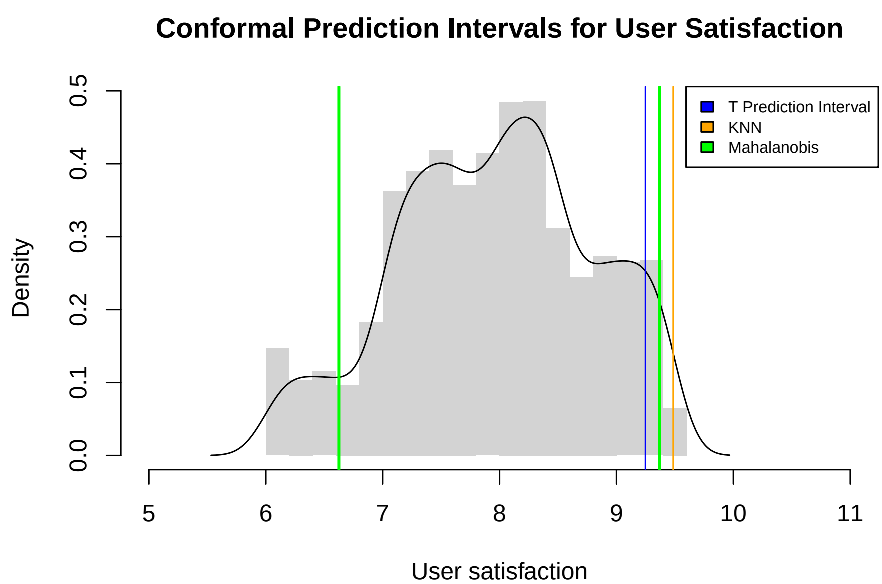


Figure 11: Conformal Prediction Intervals for User Satisfaction

All three conformal prediction methods yield very similar prediction intervals for user satisfaction, with a common lower bound around 6.6 and upper bounds between approximately 9.25 and 9.49. This consistency suggests that the results are robust to the choice of the nonconformity measure. The KNN-based interval is slightly wider, reflecting its greater sensitivity to local data density, while the T-based and Mahalanobis intervals are slightly tighter. Overall, these intervals indicate a relatively high expected level of user satisfaction for a new device, although with non-negligible uncertainty.

Having established robust prediction intervals for user satisfaction, we next examine which device features meaningfully influence these ratings through a refined modeling approach.

To facilitate the conformal prediction procedure, we first split the dataset into a training set and a calibration set, using 25% of the data for calibration and the remaining 75% for model training.

Starting from the full GAM including all technical, market and brand covariates, we performed a sequence of permutation-based significance tests, using the Freedman and Lane scheme, to assess the contribution of each predictor. Since the residual diagnostics indicated departures from normality, classical parametric tests on smooth terms could not be reliably interpreted.

Variables whose effects were not statistically significant under this permutation procedure were removed from the model. This iterative testing process led from the initial full specification to a more parsimonious final model that retains only predictors with statistically supported explanatory power for user satisfaction, ensuring that the final structure reflects genuine predictive relationships rather than noise or distributional artifacts. The final model is the following:

$$\text{User Satisfaction}_i = \beta_0 + f(\text{Price}_i) + \sum_{b=1}^{4} \beta_b \, I\{\text{Category}_i = b\} + \varepsilon_i$$

where $i$ is the device index.

This model provides a parsimonious yet powerful representation of user satisfaction by combining a flexible nonlinear effect of price with systematic category-specific shifts. The smooth term for price indicates a strong, nonlinear relationship between cost and perceived satisfaction, while the category coefficients capture structural differences in user expectations and experience across device types.

From a real-world perspective this model makes perfect sense. Indeed, price works as a rough summary of many things people care about but that are hard to measure directly, like how well the device is built, how good it looks, how reliable the software is, how much users trust the brand, the quality of customer support and how well the device fits into the rest of the company's apps and services. At the same time, device category reflects fundamentally different use cases and user expectations: a smartwatch, a fitness tracker and a smart ring are evaluated according to different criteria and daily interaction patterns. Satisfaction is therefore not only about "how good" a device is in absolute terms, but about how well it fulfills the typical role of its category. Together, price and category summarize the dominant behavioral and market-level drivers of satisfaction, making them realistic and sufficient predictors, even without explicitly modeling every individual technical specification.

The second part of the split conformal prediction procedure starts from the fitted GAM and uses the calibration set to quantify model uncertainty in a data-driven way. First, predictions are generated on the calibration data and non-conformity scores are computed as the absolute residuals between observed and predicted user satisfaction. These residuals represent how wrong the model typically is when predicting unseen data. The empirical $(1 - \alpha)$-quantile of these scores, $\hat{q}$, defines the error tolerance level needed to achieve the desired coverage probability.

This tolerance is then applied to new predictions: for every price and category combination in the prediction grid, the conformal interval is constructed as $[\hat{y}(x_{new}) - \hat{q},\ \hat{y}(x_{new}) + \hat{q}]$, where $\hat{y}(x)$ is the GAM prediction. The resulting intervals are shown in Figure 12.

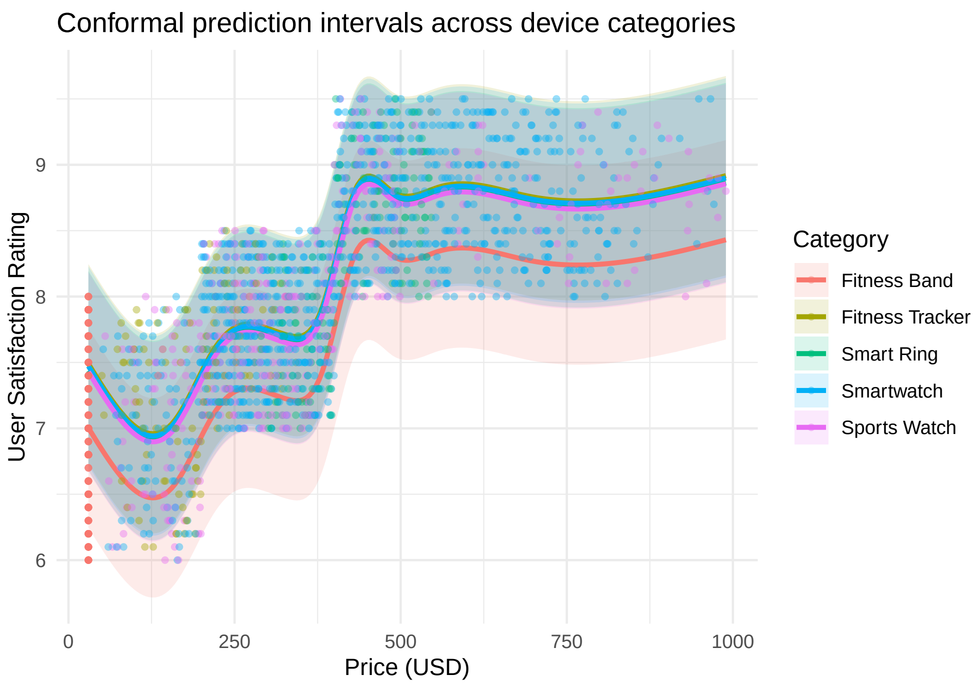


Figure 12: Split Conformal Prediction Intervals across device categories

The visualization of the conformal prediction intervals confirms the structural dynamics described in our model, specifically highlighting how the nonlinear influence of price and the discrete shifts between categories govern user satisfaction. As shown in the plot, satisfaction scores exhibit a distinct step behavior, where a sharp increase occurs between the 250 USD and 500 USD price points before stabilizing, suggesting that higher price tiers act as a reliable proxy for the premium build quality and ecosystem integration discussed previously. The overlapping yet distinct intervals for each category illustrate the systematic shifts captured by the $\beta_b$ coefficients; for instance, the Fitness Band category consistently occupies the lower bound of satisfaction across the price spectrum, while all the others maintain higher baseline expectations. This visual representation reinforces the idea that while price serves as a powerful summary of technical and brand value, the device category remains a fundamental lens through which users calibrate their satisfaction.

While split conformal prediction provides valid uncertainty quantification, it implicitly assumes that the magnitude of prediction errors is roughly constant across the covariate space. However, both the exploratory analysis and the fitted GAM suggest that the variability of user satisfaction is heteroskedastic, changing systematically with price and device category. To account for this structure, we extend the conformal framework using normalized conformal prediction, which adapts interval width to local uncertainty levels.

Specifically, after fitting the final GAM on the training set, we model the absolute residuals as a function of price and category. Residuals are log-transformed and a second GAM is fitted to capture how prediction uncertainty varies across the feature space. This model provides an estimate of a local scale parameter, $\hat{\sigma}(x)$, representing the expected magnitude of prediction error at a given price–category combination. Nonconformity scores are then computed by normalizing calibration residuals by their corresponding $\hat{\sigma}(x)$ and the $(1-\alpha)$-quantile of these normalized scores, $\hat{q}_{\text{norm}}$, determines the global tolerance level.

Prediction intervals for new devices are constructed as $[\hat{y}(x_{new}) - \hat{q}_{\text{norm}}\hat{\sigma}(x_{new}),\ \hat{y}(x_{new}) + \hat{q}_{\text{norm}}\hat{\sigma}(x_{new})]$, yielding adaptive intervals that widen in regions of high uncertainty and shrink where the model is more reliable. The resulting normalized conformal bands (Figure 13) preserve the finite-sample coverage guarantees of conformal prediction while providing a more realistic and informative representation of uncertainty, especially across heterogeneous price ranges and device categories.

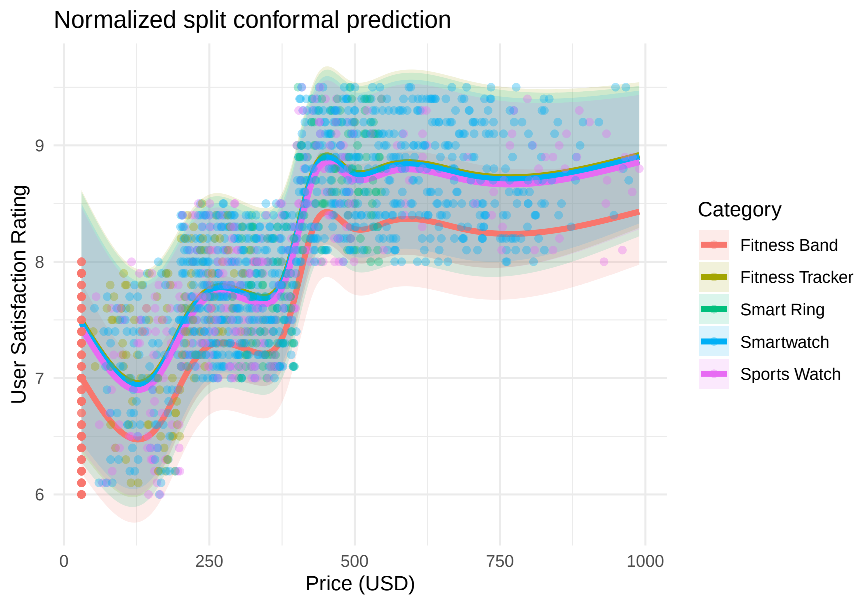


Figure 13: Normalized Conformal Prediction Intervals

In Figure 14, we visualize the normalized conformal prediction intervals for each category. The width of the prediction intervals varies along the price axis, becoming wider in regions where data are sparse or variability is higher, particularly at the lower and upper ends of the price range. This effect is especially visible for categories such as Smart Rings and Fitness Bands, where fewer observations lead to greater uncertainty. In contrast, categories with denser price coverage, such as Smartwatches and Sports Watches, exhibit tighter intervals over most of the observed range, reflecting more stable and predictable satisfaction patterns. Overall, the figure highlights how both price dynamics and category-specific usage contexts shape not only expected user satisfaction but also the reliability of predictions for new devices.

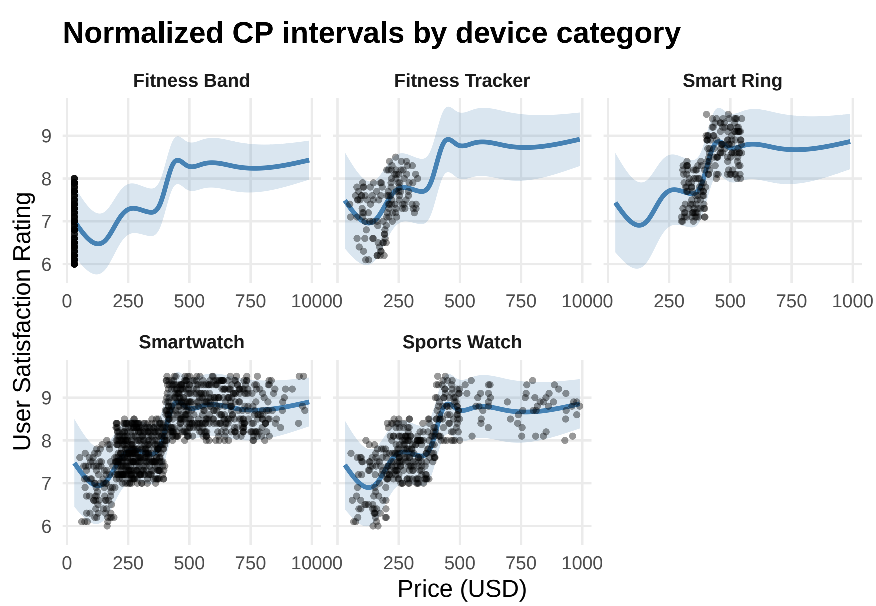


Figure 14: Normalized CP Intervals across device categories

To complement the global analysis, we conclude by illustrating how the proposed modeling and normalized conformal prediction framework can be used to produce device-specific predictions for a concrete, hypothetical product. Fixing the price at 700 USD and considering the same technical specifications, we evaluated the expected user satisfaction for two alternative category assignments: Smart Ring and Smartwatch. In both cases, the GAM yields a high point prediction, around 8.7–8.8, indicating that a device at this price point is expected to achieve strong user satisfaction regardless of the exact category. However, the prediction intervals reveal subtle but meaningful differences. The Smart Ring configuration leads to a slightly lower predicted satisfaction (8.70) and a wider upper range, with a normalized conformal interval spanning approximately [7.77, 9.63], reflecting greater uncertainty consistent with the higher variability observed in this category. In contrast, the Smartwatch prediction is marginally higher (8.76) and more tightly bounded, with an interval of roughly [7.93, 9.58], indicating more stable and predictable satisfaction outcomes. This comparison highlights how, even for identical prices and specifications, category-specific effects influence both expected satisfaction and uncertainty, reinforcing the importance of incorporating structural market differences when predicting the performance of new devices.

# Conclusion

This study provides a comprehensive, nonparametric assessment of performance, pricing and user satisfaction in the wearable health device market in the USA from a customer-centered perspective. Addressing the first research question, we find that device categories differ substantially in how they balance accuracy, battery life and technological features. Smartwatches emerge as the most balanced option for typical users, offering strong overall performance through rich sensor sets, high tracking accuracy and broad connectivity, while Sports Watches closely follow by prioritizing battery endurance and robustness. In contrast, Fitness Trackers and especially Fitness Bands exhibit more limited performance profiles, reflecting their simpler design and narrower use cases.

Regarding the second research question, our results show that higher prices are only weakly explained by objective improvements in accuracy and functionality once brand identity is taken into account. While a positive performance–price relationship exists in isolation, brand effects dominate price formation, particularly for well-established manufacturers, indicating that customers often pay a substantial premium for brand value rather than proportional technical gains.

Finally, in addressing the third research question, we demonstrate that user satisfaction for new devices can be predicted reliably using a combination of price and device category, with conformal prediction providing meaningful, distribution-free uncertainty quantification. The results highlight that satisfaction is shaped less by individual technical specifications but more by price and category-specific user expectations.

From a real-world perspective, these findings provide actionable guidance for potential customers navigating a crowded and often confusing wearable market. By quantifying how performance, price and satisfaction vary across device categories, the analysis helps users align their purchase with their actual needs rather than relying on marketing claims or brand reputation alone. In particular, the results suggest that paying a higher price does not always translate into proportionally higher technical accuracy or user satisfaction, especially once category and brand effects are considered. This insight can help consumers avoid overpaying for marginal improvements and instead select devices that offer the best balance between functionality, comfort and expected satisfaction within their intended use case.

Looking ahead, future developments could further enhance consumer decision-making by incorporating longitudinal usage data, modeling how brand perception evolves over time and integrating post-purchase behavior such as device abandonment or engagement, allowing satisfaction predictions to reflect not only initial impressions but also long-term user experience.

# References


- Kaggle, *Wearable Health Device Performance Data 2025.* https://www.kaggle.com/datasets/pratyushpuri/wearable-health-devices-performance-analysis
- market.us, Feb 2026. *Global Wearable Fitness Trackers Market* https://market.us/report/global-wearable-fitness-trackers-market/#overview
- Rachael Schultz, Steven Cohen, 2026. *The best fitness trackers and wearables of 2026.* https://www.businessinsider.com/guides/tech/best-fitness-trackers#how-we-test